

Connecting Communities: Libraries and the Power of Open Government Data

Introduction

Data is the cornerstone of today's digital society, and government agencies hold large volumes of it. This wealth of information represents an untapped source of value for both the public and private sectors. *Open government data (OGD)* refers to data collected by government entities made freely accessible to the public for use, reuse, and redistribution (Townsdin, 2018). The exact number of open government data libraries available globally can be challenging to pinpoint; however, domestically, there are over 300,000 datasets available for public use (U.S. General Services Administration, n.d.). These datasets are made available to create a more "transparent, participatory, and collaborative" government where citizens feel involved. Citizens can monitor government activities, identify areas for improvement, highlight issues that may have gone unnoticed by the government, and create tools to improve quality of life (Townsdin, 2018).

While the number of public entities disclosing their data has grown significantly, achieving the full potential of open government data remains a significant challenge. This is particularly true in ensuring its effective publication and accessibility for all interested stakeholders. Simply publishing the data only satisfies the requirement of making it available to the public and fails to ensure it is findable or usable. Recognizing that data holds value only when utilized, six key challenges have been identified to hinder the initiative from achieving its full potential: data formatting, ambiguity, discoverability, representation, overlapping scopes, and limited public participation (Attard, Orlandi, Scerri, & Auer, 2015). *Public participation* is a challenge and a vital component of the overall feedback loop the initiative hopes to create. Enhanced stakeholder participation and collaboration would enable the government to offer more citizen-centered services and data. A notable gap exists between the data and the stakeholders intended to utilize it.

Libraries play a vital role in connecting open data with the community by focusing on accessibility and community needs. As trusted information centers, they can curate, interpret, and share open government data in relevant and meaningful ways with residents. Through training, workshops, and user-friendly tools, libraries empower individuals—including students, job seekers, entrepreneurs, and civic leaders—to make informed decisions based on reliable data. This intermediary role enhances the practical value of open government data and strengthens the relationship between public institutions and the communities they serve.

This paper examines the primary types of open government data most suitable for libraries, the strategies for effectively utilizing this data, and the challenges and considerations libraries should address.

Types of Open Government Data for Libraries

The digital world continues to evolve, and libraries are crucial in ensuring people can access the information they need. By prioritizing key datasets, libraries can address urgent local needs while empowering individuals with knowledge and tools for civic, economic, health, and educational engagement. This empowerment is a key function of libraries and can significantly impact their communities.

Civic Engagement Data

Civic engagement data refers to datasets that cultivate informed and engaged constituents. Providing access to datasets on government operations, decision-making processes, and public services helps libraries enable individuals to take a more active role in their communities. Citizens can stay updated on local, state, and national issues, empowering them to make meaningful contributions. Examples include voting records, legislative updates, election data, and public spending and budget data.

As a tax-paying resident of Prince George's County, I was curious to investigate the available data on local government spending—specifically, how much each department spent, what the funds were allocated for, and who received the payments. According to the data, most of the county's funds are allocated to Public Works & Transportation, with the Department of Health trailing far behind in second place. Analyzing spending by payee revealed two prominent recipients—Stantec Consulting and MB Staffing LLC—each receiving over \$900,000 in payments from the county, significantly outpacing other payees (Prince George's County Government, 2024). This information raised several questions regarding the general allocation of funds, departmental spending, payee specifics, and future expenditures. For instance, are there systems in place that allow constituents to provide input or raise concerns about how funds are utilized? Additionally, is there an audit process to ensure that the spending is managed responsibly?

Community development plans and zoning maps are other examples of impactful data for constituents. Open access to urban planning documents and infrastructure projects can significantly benefit those who live in affected local neighborhoods. Libraries can act as facilitators by transforming complex civic datasets into actionable resources for patrons, promoting participation in local decision-making, and increasing engagement in public hearings or voting polls.

Economic and Business Data

Economic and business data is essential for driving financial growth, aiding entrepreneurs, and empowering job seekers in a community. By offering access to this data, libraries enable patrons to explore economic opportunities, analyze market trends, and make well-informed financial choices. This data type supports individual patrons and enhances local economies by promoting innovation and investment. Examples include local economic indicators such as unemployment rates and house market trends, government contracts and procurement data, labor market data, and demographic and consumer trends.

Entrepreneurs in the community can use available data to identify gaps in the market and develop business plans based on those insights. Job-seekers can make data-informed decisions about career paths, education, and training programs based on current labor market demands. Workforce data helps patrons identify industries with high earning potential and long-term job security. Local leaders and organizations can leverage economic data to advocate for investments that improve residents' quality of life but attract businesses that align with their needs.

Libraries in rural and urban areas can act as epicenters for change by making these datasets more easily digestible and accessible to residents. “The rules of engagement in economic development are changing,” said Martin Gómez, president of the Urban Libraries Council, in a recent report. “More and more, economic development success strategies involve people, technology, and growing an infrastructure for economic activity built on ideas, knowledge, experience and quality of life.” (PressReader, n.d.)

Public Health Data

Public health data helps inform communities about health trends, risks, and resources. Libraries can connect patrons with this data to improve health literacy, support informed decision-making, and create healthier communities. From disease prevention to environmental quality, public health data empowers individuals and organizations to take proactive measures that benefit personal and public well-being.

Public health data covers various topics that can address multiple community needs. These topics include statistics on infectious diseases, localized data on vaccination rates, performance information on local providers, and health outcomes categorized by age, gender, ethnicity, or socioeconomic status. Policymakers and public health practitioners rarely consider public libraries part of the health system, although they

possess several characteristics that suggest unrealized potential to advance population health. With an estimated 17,000 public libraries nationwide and four million visits each day, libraries have extensive population reach. Additionally, libraries are widely trusted by the public, making them the ideal space for coordinating and delivering health-promoting services (Philbin, Parker, Flaherty, & Hirsch, 2019).

Individuals and the broader community can benefit significantly from increased access to public health data. Improved health literacy will give patrons a better understanding of health risks, resources, and strategies for improving personal well-being. Libraries can demystify complex health statistics and terminology, making this information more accessible. The accessibility of health information has changed the decision-making process for patients concerning their health. Individuals can use data to make proactive health decisions and advocate for improved healthcare access. Data on disease outbreaks, pollution levels, or disaster preparedness can help communities enhance public safety and respond quickly to emergencies. By providing targeted resources and data for underserved populations, libraries can help address healthcare access and outcomes gaps.

Education Data

Education data provides students, parents, and educators valuable insights into academic opportunities, school performance, and educational outcomes. By connecting individuals to information on schools, colleges, financial aid, and academic success, libraries promote personal growth and contribute to improving education in the community. Access to these datasets offers a dual benefit to the community libraries, allowing them to tailor their programs and services based on their insights.

Students and parents can make decisions based on performance data on local and national schools, including graduation rates, college rankings, admission statistics, and tuition costs. Community partners can design programming for underserved and minority populations based on insights into disparities in academic performance. Educational trends and policy data surrounding funding, changes, and statistics at the national or state level can help constituents hold decision-makers accountable for positive outcomes that affect their community.

Libraries can bridge gaps in educational access by providing underserved communities with the tools and resources they need to succeed. Data-driven programs can ensure that all students have an equal opportunity for academic success. This would help create a more equitable education landscape and even the playing field for underserved communities.

By focusing on these key areas, libraries can create tailored services and programs to make open government data accessible, understandable, and actionable for all patrons. Aligning with community needs surrounding health, education, the workforce, and civic engagement enhances the library's role as a modern-day information hub.

Strategies for Libraries to Leverage Open Government Data

Libraries must be intentional and thoughtful when embedding open data into their programming and services. To unlock their full potential, libraries should adopt strategies aligned with their mission as community resources. These strategies should focus on making the data user-friendly, actionable, and inclusive for a diverse range of patrons.

Providing access to open data and training for practical interpretation is the most crucial step. Organizing workshops for users on navigating open data platforms such as Data.gov and local government data portals can enhance their confidence in finding essential datasets. Specifically, locations could organize topic-based literacy days to teach patrons how to access and interpret the metrics they see. This includes legislative, market trends, and local school performance data. Collaborating with civic tech organizations to offer practical training on data literacy and visualization tools like Tableau or Google Data Studio will motivate users to tailor and analyze datasets for fresh insights and understandings. This will encourage citizens to access and analyze data as it becomes available independently. Libraries can reduce the knowledge barrier for those unfamiliar with digital tools or government databases.

By **developing curated resources**, libraries can build data collections tailored to local priorities and interests. Patrons could also save time spent finding and organizing relevant data. Examples of this include compiled datasets and thematic knowledge bases. Similarly, **creating data visualizations** helps users save time finding relevant data by simplifying complex information, making it easier to understand and actionable. Creating interactive maps showing local trends in pandemic outbreaks or crime statistics could serve as valuable resources in local communities. Libraries could leverage tech-savvy volunteers and local universities to design infographics for the public to engage visual learners or those with lower data literacy.

Integrating data into existing library services could maximize their impact on the communities they serve. This would make the data relevant by linking it to the communities' immediate needs and enhancing the value of existing library services. An example would be incorporating labor market data into career counseling programs to help job seekers identify high-demand industries. Public health statistics could enhance scheduled wellness workshops, while civic data could elevate community forums and town hall discussions.

Each strategy requires libraries to leverage community partnerships by collaborating with local governments and organizations to maximize the use of open government data. Through these efforts, libraries can begin to break down the barriers surrounding open government data and help it reach its full potential.

Expected Challenges

Public libraries of all sizes have recognized the importance of open data; many are already active in or intend to become active in open data activities. In 2021, the Open Literacy Data Project conducted a survey to gauge the current state of open data in public libraries across Washington state. This survey aimed to answer questions surrounding interest and activity in open data, types of initiatives libraries wanted to pursue, and what they needed to make forward progress. Around 43% of respondents agreed that “**providing leadership** for local open data initiatives” should be a priority for their locations. **Technical capacity** for open data felt limited, with only two respondents agreeing that their proficiency aligned with the initiative. Across respondents, there was no consensus on which positions would most likely interact with open data, and more than half expected IT personnel to take up the responsibility. Another cited challenge involved **resources**. Many respondents (47.1%) felt it was “difficult to prioritize open data initiatives,” with funding as the common thread. For 68.6%, an “increased operating budget” was essential to progress their interests or activities in open data. **Professional development** was also a significant factor, with 56.8% feeling that more education was needed before working with the public. However, only 39.2% were interested in pursuing more data publishing, curation, and infrastructure education. (Library Journal, n.d.)

When rating the resources needed to achieve aspired open data capacity, respondents ranked “new or improved collaboration with city, state, or other partners” above the need for grant funding. Respondents believed that partnerships could help alleviate expressed **technology limitations**. The benefits of these partnerships are evident in the infrastructure support that Open Data Literacy (ODL) provides Washington state agencies. As a result, WSL supports public libraries, increasing their capabilities across the state. By building these alliances, libraries can manage resource constraints and build strengths to mobilize data, technologies, and expertise required to launch and sustain open data activities. (Library Journal, n.d.)

Conclusion

A primary critique of the Open Government Plan is its need for more engagement with the American public. As an extension of that plan, libraries can leverage open government data to increase awareness and public participation in its datasets. By

helping communities access and make sense of open government data, libraries help close the gap between complex datasets and the public.

Curating and disseminating data on core citizen interests such as civic engagement, economic opportunities, public health, and education helps libraries better serve their communities in targeted ways. Independent research into local datasets regarding county budget spending has increased my interest in civic matters that I may have otherwise ignored. With such an extensive reach, libraries' impact on public health must be recognized and capitalized on with open data. By creating dashboards and visualizations, patrons can gain pertinent information more easily.

Strategically aligning open data with their existing mission as community resources can help libraries create beneficial programming. Through workshops, interactive tools, and community partnerships, libraries address barriers such as data complexity, discoverability, and accessibility, fostering greater transparency and equity that stunts OGP's overall goal. Leadership, technical expertise, and resources are some of the challenges public libraries have encountered while pursuing open data activities. However, as evidenced in Open Data Literacy's work, we can help alleviate these challenges and better support our state libraries by increasing statewide partnerships.

As we look forward, we must shift our outlook on public libraries as just "places to read books" or "surf the Internet"; they are becoming much more than that. Libraries' evolving role as mediators of open data will continue to enhance their value in society.

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